

01 Objective

Develop a **multimodal AI system** capable of identifying potential synthetic media, impersonation and manipulated digital content, and providing an **explainable assessment** of its authenticity and associated risk.

02 Background

Generative AI has made it easy to create convincing synthetic images, video, voice and text. This becomes a serious social problem when manipulated content is used for:

Financial fraud Identity impersonation False emergency requests Reputation attacks Misleading public communication

Social engineering Fabricated evidence Coordinated misinformation

Synthetic content is no longer limited to obviously manipulated images. A scam may combine a **real person's photograph**, an **AI-generated voice**, a **fabricated document** and a **convincing conversation** into a believable identity. The Government of India has explicitly recognised AI-generated synthetic audio, video and text as a deepfake-related risk.

THE QUESTION IS NOT “IS THIS IMAGE AI-GENERATED?”

“Can an AI system examine multiple pieces of evidence and determine whether the identity, media and surrounding context are consistent and trustworthy?”

03 Suggested Pipeline



EVIDENCE SOURCES / APPROACHES YOU MAY EXPLORE

Images Video Audio Text conversations Documents URLs Metadata Identity information

Deepfake detection Speaker & face embeddings Multimodal foundation models Semantic similarity

Retrieval-augmented verification Graph-based identity analysis Anomaly detection Confidence calibration

04 Expected Deliverables

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| <ul style="list-style-type: none"> ☐ Multimodal Input
Analyse at least two evidence types ☐ Cross-Modal Verification
Compare media against identity signals ☐ Evidence-Based Explanation
Show which signals drove the verdict ☐ False-Positive Handling
Deal with uncertain or legitimate content ☐ Working Demonstration
Media → analysis → risk → verification | <ul style="list-style-type: none"> ☐ Synthetic-Media Analysis
Identify manipulation or AI generation ☐ Risk Assessment
An authenticity or impersonation-risk score ☐ Contextual Analysis
Judge context, not media in isolation ☐ Verification Workflow
Let the user independently verify a claim |
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⚠ NO SCORE IS PROOF

Teams must demonstrate that the system **does not treat an AI-generated score as absolute proof**, and provides a path for **human or independent verification**. Prioritise robustness, explainability, privacy and uncertainty awareness.

EXPECTED OUTCOME

A functional **multimodal digital-trust prototype** that helps users identify potentially manipulated media and impersonation attempts **before acting** on misleading information.